Hints: Inequalities – 3. Integral inequalities: the working techniques

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Chebyshev: if f, g are similarly ordered, $(f(x) - f(y))(g(x) - g(y)) \geqslant 0$ for all x, y. Integrate over $[a, b]^2$ and expand: $2(b-a)\int fg - 2\int f \int g \geqslant 0$. Oppositely ordered: the sign flips.

Hint 2. Jensen: let $m = \int wf$. A convex φ has a supporting line at m: $\varphi(t) \geqslant \varphi(m) + k(t-m)$ for all t. Put $t = f(x)$, multiply by $w(x) \geqslant 0$ and integrate; the linear term vanishes since $\int w = 1$. The two examples are $\varphi(t) = t^2$ and $\varphi(t) = e^t$ with $w \equiv 1$ on $[0, 1]$.

Hint 3. Normalisation: if both sides $L(f)$, $R(f)$ satisfy $L(\lambda f) = \lambda^d L(f)$, $R(\lambda f) = \lambda^d R(f)$ for $\lambda > 0$, then $L(f) \leqslant R(f)$ for all f follows from the case $\int f = 1$ (or $\|f\| = 1$, etc.): apply it to $f / \int f$ (the case $\int f = 0$ being treated separately, if needed).

Problem 2

Hint 1. The weight $(n+1)x^n$ has integral 1 on $[0, 1]$ and is increasing, like f. Which theorem compares $\int fg$ with $\int f \int g$?

Hint 2. Chebyshev with $g(x) = (n+1)x^n$: $\int_0^1 f \cdot (n+1)x^n \geqslant \int_0^1 f \cdot \int_0^1 (n+1)x^n = \int_0^1 f$.

Hint 3. Equality means $\iint (f(x) - f(y))(x^n - y^n)\, dx\, dy = 0$ with a continuous non-negative integrand, so $f(x) = f(y)$ whenever $x \neq y$: f is constant (and constants do give equality).

Problem 3

Hint 1. $\cos x$ is a weight on $[0, \pi/2]$ with values in $[0, 1]$ and total integral 1, just like the indicators of $[0, 1]$ and $[\pi/2 - 1, \pi/2]$. For a decreasing f, putting the mass as far left as possible maximises $\int f \cdot (\text{weight})$.

Hint 2. Right inequality: $\int_0^1 f - \int_0^{\pi/2} f \cos = \int_0^1 f(1 - \cos x)\, dx - \int_1^{\pi/2} f \cos x\, dx$. Bound $f \geqslant f(1)$ on $[0, 1]$ and $f \leqslant f(1)$ on $[1, \pi/2]$; the two weights have equal integrals $1 - \sin 1$.

Hint 3. Left inequality: the same with the point $c = \pi/2 - 1$, where $\int_0^c \cos = \sin c = \int_c^{\pi/2} (1 - \cos)$. Equality forces $f \equiv f(1)$ (resp. $f \equiv f(c)$) wherever the weights are positive, i.e. f constant; for strictly decreasing f both inequalities are strict.

Problem 4

Hint 1. The constraint involves x and y symmetrically and the answer contains π: substitute $x = \sin\theta$.

Hint 2. $I = \int_0^1 f = \int_0^{\pi/2} f(\sin\theta)\cos\theta\, d\theta = \int_0^{\pi/2} f(\cos\theta)\sin\theta\, d\theta$. Add the two expressions and apply the hypothesis with $x = \sin\theta$, $y = \cos\theta$: $2I \leqslant \frac{\pi}{2}$.

Hint 3. Equality: $f(x) = \sqrt{1 - x^2}$, for which $xf(y) + yf(x) = \sin(\alpha + \beta) \leqslant 1$ with $x = \sin\alpha$, $y = \sin\beta$.

Problem 5

Hint 1. Introduce a parameter: compare $F(t) = \big(\int_0^t f\big)^2 - \int_0^t f^3$ with $F(0) = 0$, and show $F' \geqslant 0$.

Hint 2. $F'(t) = f(t)\big(2\int_0^t f - f(t)^2\big)$, and $f > 0$ on $(0, 1]$ since $f' > 0$, $f(0) = 0$. Let $G(t) = 2\int_0^t f - f(t)^2$, $G(0) = 0$.

Hint 3. $G' = 2f(1 - f') \geqslant 0$ because $f' \leqslant 1$. Hence $G \geqslant 0$, so $F' \geqslant 0$ and $F(1) \geqslant 0$.